

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 05: Tuesday 25 August 2026

(Week 3)

Remark. Q04.12 to Q04.15 are leftover exercises from the Friday class in Week 2.

Remark. Q05.1 to Q05.5 are on Section 11.5 of Thomas' Calculus.

Q05.6 to Q05.9 are on Section 10.3 -10.4.

Q05.10 to Q05.14 are on Section 12.1 -12.3.

Q04.12. Consider the plane with equation $x + 3y - 7z - 8 = 0$. Find a normal vector of the plane.

Q04.13. Consider the two following two planes:

$$P_1 : x + 3y - 7z - 8 = 0,$$

$$P_2 : -2x - 6y + 14z = 5.$$

Are the two planes P_1 and P_2 parallel? Justify your answer.

Q04.14. Find an equation of the plane passing through the points $\mathbf{A}(3, 3, 0)$, $\mathbf{B}(3, 0, 1)$ and $\mathbf{C}(0, 2, 1)$.

Suggestion: First find two vectors parallel to the plane, and then find a vector normal to the plane.

Q04.15. Find an equation of the plane P_1 that passes through the points $\mathbf{A} = (-1, 1, -1)$ and $\mathbf{B} = (2, 2, 1)$, and is perpendicular to the plane $P_2 : 4x - 6y + 2z = 1$.

Suggestion: First find two vectors parallel to the plane P_1 .

Q05.1. Write down parametric equation for the line in $\mathbb{R}^3$ passing through the point $(3, -1, 4)$ and parallel to the vector $2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$. Write down also a vector equation for the line.

Q05.2. Find parametric equations for the line in $\mathbb{R}^3$ through the point $P(4, 2, 1)$ and perpendicular to two lines ℓ_1 and ℓ_2 given by:

$$\ell_1 : x = 3t + 2, y = -4t - 1, z = 4t - 9.$$

$$\ell_2 : x = 3t - 4, y = -t + 6, z = 5t.$$

Q05.3. Let $\mathbf{A} = (2, -1, 3)$, $\mathbf{B} = (4, 1, 2)$ and ℓ be the line with parametric equation

$$\mathbf{x} = 3t + 1, \quad \mathbf{y} = t + 1, \quad \mathbf{z} = -t + 2.$$

- (a) Find an equation of the plane containing $\mathbf{A}$ and ℓ .
- (b) Show that there is no plane that contains $\mathbf{A}$, $\mathbf{B}$ and the line $\ell : \mathbf{x} = 3t + 1, \quad \mathbf{y} = t + 1, \quad \mathbf{z} = -t + 2$.

Q05.4. Find parametric equations of the line through the point $(0, 1, 2)$ that is parallel to the plane $x + y + z = 2$ and perpendicular to the line $x = 1 + t, y = 1 - t, z = 2t$. Write down also a vector equation for the line.

Q05.5. Find an equation of the plane that contains the point $(-1, 2, 1)$ and the line of intersection of the planes $x + y - z = 2$ and $2x - y + 3z = 1$.

Q05.6. (a) Consider the point $P(2, -2\sqrt{3})$ in Cartesian coordinates. Rewrite the point P in polar coordinates. If possible, give an alternative answer using another ordered pair of numbers.

(b) Consider the point $Q(6, \frac{3\pi}{4})$ in polar coordinates. Rewrite the point Q in Cartesian coordinates.

Q05.7. For the following equations in Cartesian coordinates, rewrite them in terms of polar coordinates:

(i) $\frac{x^2}{9} + \frac{y^2}{4} = 1.$

(ii) $y^2 = 6x.$

Q05.8. A point $\mathbf{P}$ in $\mathbb{R}^2$ is such that both the Cartesian coordinates and polar coordinates of $\mathbf{P}$ are given by the same ordered pair of numbers. Determine the locus of $\mathbf{P}$. First give your answer in terms of polar coordinates, and then give also your answer in terms of Cartesian coordinates.

Q05.9. (a) For each part below, sketch the curve defined by the given polar equation (i.e. equation in polar coordinates):

(i) $r = 2(1 - \cos \theta)$;

(ii) $r = 6 \sin \theta$.

(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

Q05.10. Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

(ii) Evaluate the indefinite integral $\int \mathbf{v}(t) dt$.

(iii) Evaluate the definite integral $\int_0^1 \mathbf{v}(t) dt$.

Q05.11. Consider the curve in $\mathbb{R}^3$ given by

$\mathbf{r}(t) = (\ln t)\mathbf{i} + \frac{t-1}{t+3}\mathbf{j} + (t \ln t)\mathbf{k}$. Find parametric equations of the tangent line to the curve at the point where $t = 1$.

Q05.12. A particle moves in $\mathbb{R}^3$ such that its velocity at time t is given by

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

If the position of the particle at time $t = 0$ is $\mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}$. Find the position of the particle at time $t = 1$.

[Remark: The position $\mathbf{r}(t)$ and velocity $\mathbf{v}(t)$ (as vector functions of time t) of the particle satisfy $\mathbf{r}'(t) = \mathbf{v}(t)$.]

Q05.13. Two particles move in $\mathbb{R}^3$ such that their positions at time t are given by:

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

(a) Do the particles collide? Justify your answer.

(b) Do their paths intersect? Justify your answer.

Q05.14. Find the arc length of the curve

$$x = 2t - 2 \sin t, \quad y = 2 - \sqrt{2} \cos t, \quad z = \sqrt{2} \cos t$$

for $0 \leq t \leq 4\pi$.